

DINESH REDLA

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Professional Summary

Results-driven Data Science professional with 1+ years of experience in data analysis, machine learning, data preprocessing, and model development. Skilled in Python, SQL, Pandas, NumPy, Scikit-learn, and data visualization tools for extracting insights and building data-driven solutions. Experienced in handling real-world datasets, performing exploratory data analysis (EDA), feature engineering, and developing predictive models to improve business outcomes. Passionate about solving complex problems, learning new technologies, and delivering scalable AI/ML solutions.

Technical Skills

Category	Technologies
Generative AI & Agentic AI	RAG Architecture, LangGraph, LangChain, Prompt Engineering, OpenAI GPT, Claude
Machine Learning	XGBoost, LightGBM, CatBoost, Scikit-learn
NLP	Transformers (BERT, RoBERTa, T5), spaCy, NLTK
Data & Backend	Python, SQL, Pandas, NumPy
Cloud & Deployment	Microsoft Azure (Blob, Functions, AKS, OpenAI), Kubernetes
Visualization	Power BI, Matplotlib, Seaborn, Plotly

Projects

1. Document Type Classification System (Classification)

Project Description:

- Developed transformer-based document classification system achieving 96% accuracy classifying 20+ document types. Built end-to-end ML pipeline from data preprocessing through model serving handling 500K+ documents daily. Integrated comprehensive testing and validation frameworks with 15+ performance benchmarks. Reduced manual document sorting time by 85%, enabling downstream automation.

Roles & Responsibilities:

- Created labeled dataset of 150K+ documents across 20 categories with quality assurance protocols.
- Fine-tuned BERT and RoBERTa transformer models on custom document classification task .
- Implemented data augmentation strategies increasing training data effective size by 3x.
- Built document preprocessing pipeline handling PDFs, images, and text documents with OCR .
- Designed production inference system using FastAPI with batch processing and real-time serving modes .
- Containerized model using Docker and deployed on Kubernetes cluster for scalability .
- Implemented comprehensive testing framework including unit tests, integration tests, and performance benchmarks .
- Created monitoring system detecting model drift and triggering automated retraining

Project Description:

- Achieved 96% accuracy on 20-document-type classification task on validation set.
- Reduced manual document sorting effort by 85% through automated classification .
- Enabled downstream process automation processing 100K+ documents weekly .
- Deployed system with 99.8% uptime in production serving multiple business units .

2. Intelligent Document Processing Agent (Agentic AI)

Project Description:

- Built autonomous document processing agent combining computer vision, NLP, and agentic decision-making to extract structured data from unstructured documents. System intelligently handles multiple document types, validates extracted data, and escalates complex cases. Processes 100K+ documents monthly with 94% accuracy, enabling downstream ML pipelines and business intelligence.

Roles & Responsibilities:

- Designed multi-stage agentic pipeline: document preprocessing, layout analysis, entity extraction, validation, escalation.
- Integrated computer vision modules for document image enhancement, table detection, and optical character recognition .

- Built validation agent checking extracted data consistency and triggering corrections or human review
- Implemented self-correcting mechanisms allowing agents to retry failed extractions with different strategies .
- Designed escalation workflows routing complex or ambiguous documents to human operators with context .
- Created feedback loops incorporating human corrections to improve future extraction accuracy .
- Built monitoring system tracking extraction quality, error rates, and escalation patterns .
- Integrated with downstream data processing systems for automated data pipeline activation .

Key Achievements:

- Extracted structured data from 100K+ unstructured documents monthly with 94% accuracy .
- Improved data extraction accuracy from 76% baseline to 94% with iterative refinements
- Enabled 50+ downstream ML pipelines and business intelligence applications .
- Reduced manual document data entry effort by 80% through automation
- Processed documents 10x faster than manual entry with improved accuracy and consistency

Education

Sree Vidyanikethan Engineering College
B.Tech - Computer Science (Data Science)

2021 – 2025
Tirupati

Certifications

- Google – Machine Learning Crash Course
- AWS Educate – Machine Learning Foundations